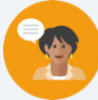


15.6 Simulating Small and Large Numbers of Trials

Lesson Introduction

 Benchmark	MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.		 Learning Targets	- I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment. - I can use results from computer-simulated events to informally define the Law of Large Numbers.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.	
 Essential Questions	- How can you use computer-simulated events to compare experimental and theoretical probabilities? - How does the number of trials in a simple experiment affect the experimental probability?			
 Lesson Narrative	This lesson is designed to give students an opportunity to use simulations to collect data, then calculate experimental probabilities and estimate theoretical probability of an event using the experimental probabilities they calculate. Different methods for conducting simulations are offered in the lesson. Some simulations use technology, while others do not. Students then practice matching situations to a simulation that could be used to determine probabilities of events.			
 Component Summary	15.6.1 Warm-Up (~5 min)	Which spinner doesn't belong? (SE pg. 427)	15.6.3 Bring It Together (10 min)	Choosing a simulation for a situation (SE pg. 429)
	15.6.2 Exploration (15–20 min)	Compare experimental results from a simulation to expected results (SE pg. 427)	15.6.4 Try It (10 min)	Matching situations with simulations to determine experimental probabilities (SE pg. 430)
	15.6.5 Wrap-Up (~5 min)	Self-reflection on the lesson (SE pg. 431)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Simulation <u>Additional Terms:</u> - Random variation - Experimental probability - Theoretical probability		N/A	
			Additional Preparation	
			(Optional) 15.6.4 Try It can be turned into task cards.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may confuse experimental and theoretical probabilities. - Students may struggle with matching a situation with a simulation that could be used to determine the experimental probability.		Consider turning the 15.6.4 Try It into a matching activity where students physically match the cards.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Think-Pair-Share In 15.6.2 Exploration, question 2c lends itself to have students engage in a Think-Pair-Share. Consider having students write down a few notes about what they notice before pairing up and comparing ideas. Partners can then share out in a class discussion what they noticed.		MA.K12.MTR.4.1: Discussion and Reflection This lesson creates many opportunities for students to discuss their thinking with peers. In 15.6.4 Try It, students work together to match the situation with the simulation that would best determine the experimental probabilities.	
 Differentiate Instruction	For a kinesthetic approach to 15.6.4 Try it, consider creating task cards out of the descriptions and simulations. Students can then physically match the descriptions to the simulations with their partner.			
Lesson Notes:				

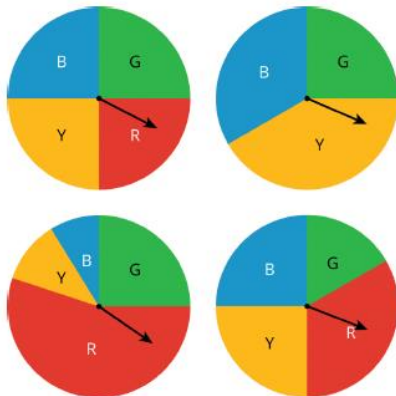
15.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students are presented with four spinners with the prompt “Which one doesn’t belong?” Students could argue that any one of the four options “doesn’t belong” for different reasons using mathematical reasoning.



15.6.1 Warm-Up: Which One Doesn't Belong?

1. Four spinners are shown.



- a. Which spinner doesn't belong?

Answers will vary. Sample answers:

- The top right spinner doesn't have red.
- The bottom left spinner is the only one with red > 50%.
- The top left spinner is the only fair spinner.
- The bottom right spinner is the only one that the green sector doesn't have a 25% of being spun.

- b. Explain your reasoning.

Answers will vary.



“Which One Doesn't Belong?” fosters a need to define terms carefully and use words precisely (probability, experiment, trials, outcome, sample space, simulation ...) in order to compare and contrast the group of spinners. Consider prompting students to include at least one term learned in this unit in their reasoning.

15.6.2 Exploration: Comparing Experimental Results to Expected Results

Component Overview: Students engage in an Exploration where they look at the results of different numbers of trials on a simulation of spinning a color spinner. Students start by finding the theoretical probability of landing on each color and compare that to the experimental probabilities from the simulation.



15.6.2 Exploration: Comparing Experimental Results to Expected Results

Work with your partner to complete the following.

1. When performing the simple experiment of flipping a coin over repeated trials, imagine each of the resulting situations listed in the table.
 - a. Complete the table to describe the results.

Situation	Is the result surprising?		Is the result possible?	
	Yes	No	Yes	No
You flip the coin once, and it lands heads-up.	☐	☒	☒	☐
You flip the coin twice, and it lands heads-up both times.	☐	☒	☒	☐
You flip the coin 100 times, and it lands heads-up all 100 times.	☒	☐	☒	☐

- b. If you flip the coin 100 times, explain how many times you would expect the coin to land heads-up.
Answers will vary. But you would expect about half of the trials to land face up.
 - c. With your partner, discuss some other results that would not be surprising if you flip the coin 100 times.
Answers will vary. Answers will be around the number 50.



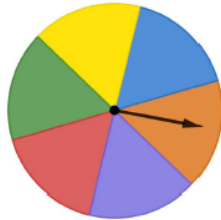
- Would you expect to get the exact same result each time?
- Why or why not?



Students can use the simulation applet for flipping a coin.

15.6.2 Exploration: Comparing Experimental Results to Expected Results (continued)

2. A spinner is shown.



- a. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Red	Purple	Orange	Blue	Yellow	Green
Theoretical Probability	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%

- b. Sara used a computer-simulated spinner to perform different numbers of trials with the spinner. The results show the percentage of times it landed on each color in each simulation.

Circle the experimental probabilities that are more than 1% different from the theoretical probability.

Color	Red	Purple	Orange	Blue	Yellow	Green
Simulation A (6 spins)	0%	16.7%	16.7%	33.3%	0%	33.3%
Simulation B (60 spins)	16.7%	15%	20%	13.3%	10%	25%
Simulation C (600 spins)	17.7%	19%	18%	15.8%	14.5%	15%
Simulation D (6000 spins)	16.9%	16%	17.1%	16.6%	16.7%	17.1%



Consider a reminder of how to find theoretical probability.

$$\frac{\text{number of desired outcomes}}{\text{total number of possible outcomes}}$$

Consider posting this formula on the board for students to reference.



- Is this simulation method free of bias?



Notice and Wonder

- What do you notice about how many experimental probabilities you circled in each simulation?
- What do you wonder about why this is?

15.6.2 Exploration: Comparing Experimental Results to Expected Results (continued)

- c. Discuss with your partner what you notice when comparing the experimental probabilities to the theoretical probabilities.
Answers will vary.
- d. In which simulation(s) were the experimental probabilities the most different from the theoretical probabilities?
Simulation A was most different from the theoretical probability.
- e. In which simulation(s) were the experimental probabilities the most similar to the theoretical probabilities?
Simulation D was most similar to the theoretical probability.
- f. Imagine Sara used the computer-simulated spinner to perform 600,000 spins. Describe how you think the experimental probabilities of landing on each color would compare to the theoretical probabilities.
Answers will vary. The experimental probability should end up being close to the theoretical probability the more times that the spinner is spun.



Think-Pair-Share

Question 2c lends itself to have students engage in a Think-Pair-Share. Consider having students write down a few notes about what they notice before pairing up and comparing ideas. Partners can then share out in a class discussion what was noticed.



- Why do you think this is?

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results

Component Overview: In this *Bring It Together*, students continue to look at simulation and experimental results of a simulation. Students also look at how they can determine a simulation that could be used to predict results for various chance events.



15.6.3 Bring It Together: Comparing Experimental Results to Expected Results



A **simulation** is an approximate imitation of a statistical experiment, often done with a computer program to examine the statistics of a large quantity of trials.

1. Complete the statement.

As the number of trials in a simulation of a simple

experiment increases, the ☐ experimental ☐ theoretical probability

gets ☐ closer to ☐ farther from the ☐ experimental ☐ theoretical probability.

2. Imagine flipping a coin 3 times, and it has come up heads once. The experimental probability of landing on heads after 3 trials is $\frac{1}{3}$. Discuss with your partner whether flipping the coin one more time is guaranteed to result in landing on heads so the experimental probability, $\frac{2}{4}$, will equal the theoretical probability $\frac{1}{2}$.

Answers will vary. It is not a guarantee that the coin will land on heads, but the more times that the coin is flipped, the closer the experimental probability will get to the theoretical probability.



Listen to student discussions to determine their understanding of random variation.

When debriefing this question, call on different students to share what they discussed with their partner based on what they heard.

The goal is to lead the debrief toward understanding that on each trial, the probability of landing on heads is $\frac{1}{2}$ but that due to random variation, it doesn't always work out that the coin lands on heads exactly half the time.

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results (continued)



When performing simple experiments, the experimental probabilities may differ from theoretical probabilities due to random variation. However, as the number of repetitions increases, experimental probabilities will typically better approximate the theoretical probabilities.



Consider a class discussion on what random variation means in an experiment.

Simulations are useful when it is hard or time consuming to gather enough information to estimate the probability of some event.

3. Consider the situations described in the table.

Situation	Description
A	The local meteorologist in Ocala, FL says there is a 50% chance of rain tomorrow.
B	A science teacher at Hamilton County High School puts the names of all students in a bag and randomly selects which of 6 groups they will sit in for his new seating chart.
C	The U.S. Centers for Disease Control and Prevention (CDC) released data in 2016 showing that $\frac{1}{3}$ of all adults in the US do not get enough sleep on a regular basis.



Consider asking students to highlight or underline any words or phrases which they aren't sure how to explain. This can provide insight into vocabulary or contexts that may need to be discussed to build students' understanding.

Explain which situation could be simulated using the spinner from the Exploration.

The situation that could best be used by a spinner simulation is Situation B. The spinner has 6 sections, and the science teacher has 6 groups. The spinner will provide a truly random selection.

Answers may vary as students could also make an argument for Situation A.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

Component Overview: In this Try It activity, students match four descriptions of situations to a simulation that could be used to determine the experimental probabilities of the event.



15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

- Four situations are described.

Situation	Description
A	In a small lake, 25% of the fish are female. You capture a fish, record whether it is male or female, and toss the fish back into the lake. If you repeat this process 5 times, what is the probability that at least 3 of the 5 fish are female?
B	Elena makes about 80% of her free throws. Based on her past successes with free throws, what is the probability that she will make exactly 4 out of 5 free throws in her next basketball game?
C	On a game show, a contestant must pick one of three doors. In the first round, the winning door has a vacation. In the second round, the winning door has a car. What is the probability of winning a vacation and a car?
D	Your choir is singing in 4 concerts. You and one of your classmates both learned the solo. Before each concert, there is an equal chance the choir director will select you or the other student to sing the solo. What is the probability that you will be selected to sing the solo in exactly 3 of the 4 concerts?



For a kinesthetic approach to this activity, consider creating task cards out of the descriptions and simulations. Students can then physically match the descriptions to the simulations with their partner.

Work with your partner to match each situation to a simulation on the next page.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities (continued)

Simulation	Situation
Toss a standard number cube 2 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 1 or 2 appeared both times to estimate the probability.	<i>C</i>
Make a spinner with four equal sections labeled 1, 2, 3, and 4. Spin the spinner 5 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 4 appears 3 or more times to estimate the probability.	<i>A</i>
Toss a fair coin 4 times and record the outcomes. Repeat this process many times, and find the proportion of the simulations in which exactly 3 heads appear to estimate the probability.	<i>D</i>
Place 8 blue chips and 2 red chips in a bag. Shake the bag, select a chip, record its color, and then return the chip to the bag. Repeat the process 4 more times to obtain a simulated outcome. Then repeat this process many times and find the proportion of the simulations in which exactly 4 blues are selected to estimate the probability.	<i>B</i>



- How do you think the experimental probabilities that would be collected from these simulations will compare to their theoretical probabilities?



To extend students' thinking, have students choose one or more of the simulations and conduct the experiment and record their results and the experimental probability. How does it compare to the expected theoretical probability?

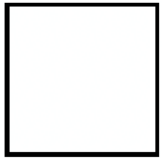
15.6.5 Wrap-Up: Reflection

Component Overview: Students self-reflect on simulating the probability of events.



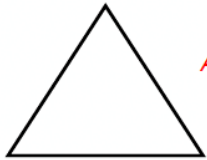
15.6.5 Wrap-Up: Reflection

1. What is something that is now “squared away” in your mind about simulating the probability of events?



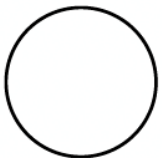
Answers will vary.

2. What are three “points” that are important about simulating the probability of events?



Answers will vary.

3. What is something that is still “circling” your brain about simulating the probability of events?



Answers will vary.



In the next lesson, students will continue to use simulations to find the probability of events. Look for common themes in what students are struggling with to address in the next lesson.